

SIMATS ENGINEERING

Assessment Tool 1 – Short Answer Test: Model Answers

Course: CSA07 – Computer Networks | CO1: Apply the functions of OSI layer in enabling reliable data transmission across networks

Question 1

Transfer a 150 MB file over a 50 Mbps link; Transport Layer segments = 1500 bytes; Data Link Layer adds 40 bytes header+trailer per frame. Find segments, frames, DLL overhead, and transmission time.

Solution:

File size = 150 MB = 150,000,000 bytes

Number of segments = $150,000,000 / 1500 = 100,000$ segments

Since each segment is encapsulated into one frame: Number of frames = **100,000 frames**

Size of each frame = $1500 + 40 = 1540$ bytes

Total Data Link Layer overhead = $100,000 \times 40$ bytes = 4,000,000 bytes = **4 MB (32,000,000 bits)**

Total data transmitted (with overhead) = $100,000 \times 1540 = 154,000,000$ bytes = 154 MB

Total bits to transmit = $154,000,000 \times 8 = 1,232,000,000$ bits

Transmission time = $1,232,000,000 \text{ bits} / 50,000,000 \text{ bps} = 24.64$ seconds

Explanation:

The Transport Layer ensures reliable communication by segmenting the file into manageable units, numbering each segment for correct sequencing, using acknowledgements to confirm receipt, and retransmitting any segment that is lost or corrupted (e.g., as done by TCP). It also performs flow and congestion control so the sender does not overwhelm the receiver or the network. The Data Link Layer ensures reliability on each physical hop by framing the data, attaching error-detection codes (such as CRC) in the trailer, performing MAC-level addressing, and requesting retransmission of damaged frames between adjacent nodes. Together, these two layers guarantee that data arrives completely, in order, and without errors from source to destination.

Question 2

A Transport Layer protocol uses a sliding window of 6 frames (1024 bytes each). Sender transmits 48 frames. Find the number of transmission windows and total retransmissions if one frame per window is lost.

Solution:

Window size = 6 frames, Total frames to send = 48

Number of transmission windows = $48 / 6 = 8$ windows

If 1 frame is lost per window, retransmissions needed = 1×8 windows = **8 retransmissions**

Explanation:

Flow control (implemented here through the sliding window mechanism) allows the sender to transmit multiple frames (a full window) before waiting for individual acknowledgements, instead of sending one frame and waiting for its acknowledgement each time. This keeps the communication channel busy, improves throughput and bandwidth utilization, and prevents a fast sender from overwhelming a slow receiver's buffer. When a frame is lost, only that frame (or window, depending on the protocol - Go-Back-N or Selective Repeat) needs retransmission, which keeps recovery efficient while still protecting the receiver from being flooded with unacknowledged data.

Question 3

A 12,000-byte IP packet is transmitted through a router with MTU = 1500 bytes. Each fragment carries a 20-byte IP header. Find the number of fragments and total header overhead.

Solution:

Maximum data (payload) per fragment = MTU – header = 1500 – 20 = 1480 bytes

Total packet size = 12,000 bytes

Number of fragments = $\text{ceil}(12,000 / 1480) = \text{ceil}(8.11) = \mathbf{9 \text{ fragments}}$

(8 fragments × 1480 bytes = 11,840 bytes; remaining 160 bytes go into the 9th fragment)

Total header overhead = 9 fragments × 20 bytes = **180 bytes**

Explanation:

The Network Layer ensures successful delivery of fragmented packets by assigning the same Identification number to all fragments of the original packet, using the Fragment Offset field to indicate each fragment's position within the original data, and using the More Fragments (MF) flag to mark all but the last fragment. Each fragment is routed independently across the network as a separate IP packet. At the destination, the Network Layer uses the identification number and offset values to reassemble the fragments in the correct order into the original packet before passing it up to the Transport Layer. If any fragment is lost, the entire original packet must be retransmitted.

Question 4

(Same as Question 2) A Transport Layer protocol uses a sliding window of 6 frames (1024 bytes each). Sender transmits 48 frames. Find the number of transmission windows and total retransmissions if one frame per window is lost.

Solution:

Window size = 6 frames, Total frames to send = 48

Number of transmission windows = $48 / 6 = \mathbf{8 \text{ windows}}$

If 1 frame is lost per window, retransmissions needed = $1 \times 8 \text{ windows} = \mathbf{8 \text{ retransmissions}}$

This question is identical to Question 2, so the working and results are the same.

Explanation:

Flow control (implemented here through the sliding window mechanism) allows the sender to transmit multiple frames (a full window) before waiting for individual acknowledgements, instead of sending one frame and waiting for its acknowledgement each time. This keeps the communication channel busy, improves throughput and bandwidth utilization, and prevents a fast sender from overwhelming a slow receiver's buffer. When a frame is lost, only that frame (or window, depending on the protocol - Go-Back-N or Selective Repeat) needs retransmission, which keeps recovery efficient while still protecting the receiver from being flooded with unacknowledged data.

Question 5

A 600 MB video file is transferred with a Session Layer checkpoint every 60 MB. Failure occurs after 390 MB. Find checkpoints created before failure and data to be retransmitted after recovery.

Solution:

Checkpoints occur at: 60, 120, 180, 240, 300, 360, 420 MB, ...

Failure occurs at 390 MB, which lies between the checkpoint at 360 MB and the next one at 420 MB

Number of checkpoints created before failure = $\text{floor}(390/60) = \mathbf{6 \text{ checkpoints}}$ (last one at 360 MB)

Data already confirmed at last checkpoint = 360 MB

Data transmitted at time of failure = 390 MB

Data to be retransmitted after recovery = $390 - 360 = 30$ MB

Explanation:

Synchronization checkpoints in the Session Layer are vital because they allow a session to resume from the last confirmed checkpoint rather than restarting the entire transfer after a failure. This significantly reduces the amount of data that must be retransmitted (only 30 MB instead of the full 390 MB already sent), saving time, bandwidth, and processing resources. The Session Layer is responsible for establishing, maintaining, synchronizing, and gracefully recovering or terminating communication sessions between two systems, which is especially important for large file or multimedia transfers prone to interruption.

Question 6

A 900 MB file is compressed by 40% at the Presentation Layer, then encrypted with 5% overhead added to the compressed size. Find the compressed size, encrypted size, and total percentage reduction.

Solution:

Compressed size = $900 \text{ MB} \times (1 - 0.40) = 900 \times 0.60 = 540 \text{ MB}$

Encryption overhead = 5% of compressed size = $540 \times 0.05 = 27 \text{ MB}$

Encrypted (final) size = $540 + 27 = 567 \text{ MB}$

Total reduction compared to original = $900 - 567 = 333 \text{ MB}$

Total percentage reduction = $(333 / 900) \times 100 = 37\%$

Explanation:

The Presentation Layer improves transmission efficiency by compressing data before it is sent, which reduces the volume of data transmitted, lowers bandwidth consumption, and shortens transmission time, even after the encryption overhead is added back (a net 37% reduction here). It improves security by encrypting the data, ensuring confidentiality and protecting it from unauthorized access or eavesdropping during transit. The Presentation Layer also handles data translation and formatting (e.g., character encoding, data structure conversion) so that data sent by one system's application can be correctly interpreted by another, regardless of differences in internal data representation.

Question 7

An FTP application transfers 3 GB of data over a link with 120 Mbps effective throughput. The Application Layer divides data into 3 MB requests. Find the total number of requests and the transmission time.

Solution:

Total data = $3 \text{ GB} = 3,000 \text{ MB} = 3,000,000,000 \text{ bytes}$

Number of requests = $3,000 \text{ MB} / 3 \text{ MB} = 1,000 \text{ requests}$

Total bits to transmit = $3,000,000,000 \text{ bytes} \times 8 = 24,000,000,000 \text{ bits}$

Transmission time = $24,000,000,000 \text{ bits} / 120,000,000 \text{ bps} = 200 \text{ seconds}$

Explanation:

The Application Layer provides reliable services to users by offering standardized protocols (such as FTP) through which user applications can request, send, and receive files without needing to manage low-level network operations. It breaks large transfers into smaller requests, which allows better error handling, progress tracking, and recovery from partial failures (only a failed request needs to be resent, not the whole file). It also relies on the underlying Transport Layer (TCP) for guaranteed, in-order, error-checked delivery, and it manages user-facing concerns such as authentication, session

management, and presenting data in a usable form to the end user.

Question 8

A packet experiences processing delays of Application = 4 ms, Transport = 3 ms, Network = 5 ms, Data Link = 2 ms, Physical = 1 ms, with propagation delay = 18 ms and transmission delay = 12 ms. Find the total end-to-end delay.

Solution:

Total processing delay = $4 + 3 + 5 + 2 + 1 = 15$ ms

Total end-to-end delay = Processing delay + Transmission delay + Propagation delay

Total end-to-end delay = $15 + 12 + 18 = \mathbf{45\ ms}$

Explanation:

Each OSI layer contributes to the total delay through the processing it performs: the Application Layer prepares and formats the user's data/request; the Transport Layer adds segmentation, sequencing and reliability information; the Network Layer performs routing and logical addressing decisions (often the largest processing delay, as seen here); the Data Link Layer performs framing, addressing, and error-detection code generation; and the Physical Layer converts the frame into raw bits for transmission. In addition to this cumulative processing delay, transmission delay (time to push all bits onto the link) and propagation delay (time for signals to travel the physical distance) add to the total end-to-end delay experienced by the packet.

Question 9

80 MB of useful data is sent using 1600-byte frames, each carrying 80 bytes of protocol overhead. Find the number of frames, total overhead, and transmission efficiency.

Solution:

Useful data per frame (payload) = $1600 - 80 = 1520$ bytes

Total useful data = 80 MB = 80,000,000 bytes

Number of frames required = $\text{ceil}(80,000,000 / 1520) = \mathbf{52,632\ frames}$

Total overhead transmitted = $52,632 \times 80\ \text{bytes} \approx \mathbf{4,210,560\ bytes\ (\approx 4.21\ MB)}$

Total data transmitted = $52,632 \times 1600\ \text{bytes} \approx 84,211,200\ \text{bytes}$

Transmission efficiency = $(\text{Useful data per frame} / \text{Total frame size}) \times 100 = (1520/1600) \times 100 = \mathbf{95\%}$

Explanation:

Protocol overhead (headers/trailers added by various OSI layers) directly reduces the effective throughput of a network because part of every frame's capacity is consumed by control information rather than actual user data. This lowers transmission efficiency (here, 95%, meaning 5% of the bandwidth is consumed purely by overhead), increases the total number of frames and total bits that must be transmitted for the same amount of useful data, and consequently increases transmission time. Network designers must balance frame size and overhead carefully - very small frames waste more bandwidth on repeated overhead, while very large frames increase the impact of a single frame error.

Question 10

A 240 MB medical record is sent over an IoT healthcare network. The Presentation Layer compresses it by 30%, the Transport Layer divides the compressed data into 1400-byte segments, and the Data Link Layer adds 60 bytes of overhead per frame. Find the compressed size, number of segments, total protocol overhead, and final data sent over the Physical Layer.

Solution:

Compressed size = $240\ \text{MB} \times (1 - 0.30) = 240 \times 0.70 = \mathbf{168\ MB}$

Compressed size in bytes = 168,000,000 bytes

Number of segments = $168,000,000 / 1400 = 120,000$ segments

Data Link Layer overhead (60 bytes/frame, one frame per segment) = $120,000 \times 60 = 7,200,000$ bytes **(7.2 MB)**

Final data transmitted across the Physical Layer = Compressed data + DLL overhead = $168,000,000 + 7,200,000 = 175,200,000$ bytes **(175.2 MB)**

Explanation:

Multiple OSI layers work together to ensure secure and reliable delivery of the medical record: the Presentation Layer compresses (and would typically also encrypt) the sensitive patient data, reducing size and protecting confidentiality; the Transport Layer divides the compressed data into fixed-size segments, adds sequencing information, and ensures reliable, in-order, error-checked delivery, retransmitting lost segments if necessary; the Network Layer provides logical addressing and routing of each segment across the IoT healthcare network to the correct destination; the Data Link Layer frames each segment, adds error-detection information, and manages access to the physical medium between devices; and the Physical Layer transmits the final bits as electrical, optical, or radio signals. This layered cooperation ensures the medical data reaches its destination completely, accurately, securely, and efficiently despite the resource constraints typical of IoT devices.